

Programme	B. Sc. Mathematics Honours			
Course Code	MAT6EJ302(1)			
Course Title	MATHEMATICS FOR DIGITAL IMAGES			
Type of Course	Elective (Specialisation- Mathematical Computing)			
Semester	VI			
Academic Level	300 - 399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Basic Geometry and Algebraic Structures			
Course Summary	The focus of this paper is mathematics underlying patterns which in converse can be used to produce patterns automatically by computer, allocating some design decisions to the user. We begin with isometries, those transformations of the plane which preserve distance and hence shape. These fall into two classes: the direct ones are rotations or translation, and the indirect ones reflections or glides. We derive the rules for combining isometries, and introduce groups, and the dihedral group in particular. We also apply this to classifying all 1-dimensional or 'braid' patterns into seven types. Our next focus is on symmetries; that is, those isometries which send a pattern onto itself, each part going to another with the same size and shape. A plane pattern is one having translation symmetries in two non-parallel directions. These are made up of parallelogram shaped cells, falling into five types. Finally, we deduce the existence of 17 pattern types, each with its own set of interacting symmetry operations.			

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe the concept of isometries in geometry, including translation, rotation, and reflection, and understand their properties and how they preserve distances.	U	C	Internal Exam/Assignment /Seminar/ Viva / End Sem Exam
CO2	Demonstrate the ability to compose isometries, understand their combined effects, and analyse the outcomes of sequential transformations.	Ap	P	Internal Exam/Assignment /Seminar/ Viva / End Sem Exam
CO3	Investigate the classification of plane patterns, including different net types such as parallelogram nets, rectangular nets, centred rectangular nets, square nets, and hexagonal nets, and analyse examples of the 17 plane pattern types.	An	F	Internal Exam/Assignment /Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book	MATHEMATICS FOR DIGITAL IMAGES : Creation, Compression, Restoration, Recognition. S G Hoggar- Cambridge University Press.			
Module	Unit	Content	Hrs (48+12)	Ext. Marks (70)
I	Introduction		12	
	1	Isometries and their sense		
	2	The plane and vectors		
	3	Isometries – Translation, Rotation, Reflection		
	4	The sense of an isometry		
	5	The Classification of isometries		
	6	Composing isometries		
	Sections from Text (i): Chapter 1 – 1.1, 1.2, 1.3			
II	How Isometries Combine		12	
	7	Reflections are the key		
	8	Some useful compositions		
	9	The Image of a line of symmetry		
	10	The dihedral group		
	11	Appendix on groups		
	Sections from Text (i):Chapter 2 – 2.1, 2.2, 2.3, 2.4, 2.5			
III	The Seven Braid Patterns, Plane Patterns & Symmetries		12	
	12	Classification of braids		
	13	Constructing braid patterns		
	14	Translations and nets		
	15	Cells		
	16	The five net types		
	17	Nets allowing a reflection		
	Sections from Text (i): Chapter 3, Chapter 4 – 4.1, 4.2, 4.3			
IV	The 17 Plane Patterns		12	
	18	Preliminaries		
	19	The general parallelogram net		
	20	The rectangular net		
	21	The centred rectangular net		
	22	The square net		
	23	The hexagonal net		
	24	Examples of the 17 plane pattern types		
	25	Scheme for identifying pattern types		

	Sections from Text (i): Chapter 5 – 5.1, 5.2, 5.3, 5.4, 5.5, 5.6, 5.7, 5.8		
V (Open Ended)	Advanced Topics		12
	26	Basic Syntax and Scalar arithmetic operations and calculations by Using MATLAB	
	27	Arithmetic operations in matrix data & Reading an Image File by Using MATLAB	
References:			
1. Baldock R and Graham J (2000) Image Processing and analysis, a practical approach, Oxford University Press 2. Gonzalez R C and Woods R E (1993) Digital Image Processing, Addison-Wesley			

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	3	2	1	3	1	2	2	3	0	2
CO 2	2	3	2	1	2	1	2	2	2	0	2
CO 3	3	3	2	1	3	1	3	3	3	0	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓